

# Erdős Problem #408

|        |                |        |               |
|--------|----------------|--------|---------------|
| EP-408 | L1 - Tractable | Ouvert | Number Theory |
|--------|----------------|--------|---------------|

## Énoncé original (anglais, reproduction mot pour mot)

Let  $\phi(n)$  be the Euler totient function and  $\phi_k(n)$  be the iterated  $\phi$  function, so that  $\phi_1(n) = \phi(n)$  and  $\phi_k(n) = \phi(\phi_{k-1}(n))$ . Let  $f(n) = \min \{k : \phi_k(n) = 1\}$ . Does  $f(n)/\log n$  have a distribution function? Is  $f(n)/\log n$  almost always constant? What can be said about the largest prime factor of  $\phi_k(n)$  when, say,  $k = \log \log n$ ?

Source et attribution. Énoncé reproduit depuis UnsolvedMath, jeu de données officiel sous licence CC BY 4.0.

Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Jeu de données: <https://huggingface.co/datasets/ulamai/UnsolvedMath>